ECE 310 Fall 2026
Lecture 22: Spectral Analysis Part 1

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Learning Objectives

After this lecture, you should be able to:

- State the DTFT spectrum of a finite-length sinusoidal signal and how the length of the signal affects the corresponding DFT.
- Define the spectral analysis problem.
- Discuss the challenges with separating multiple sinusoids present in a given finite-length signal.



Spectral Analysis Problem

Spectral analysis

Spectral analysis is a common application of digital signal processing where we seek to identify unknown sinusoidal components of a finite-length signal.

$$x[n] = \sum_{m=1}^{M} A_m \cos(\omega_m n), \ 0 \leq n \leq N-1$$

Unknowns include:

- Number of spectral components M.
- Frequencies ω_m, $1 \leq m \leq M-1$.
- Amplitudes A_m, $1 \leq m \leq M-1$.

If $N \to \infty$, this is trivial because we just locate Dirac deltas.



DTFT of Finite-length Sinusoid

Let $x[n] = A\cos(\omega_0 n),\ 0 \le n \le N-1$. Then,

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} A\cos(\omega_0 n)e^{-j\omega n} = \frac{A}{2}\sum_{n=0}^{N-1} e^{-j(\omega-\omega_0)n} + e^{-j(\omega+\omega_0)n}$$

$$= \frac{A}{2}\left[\frac{1-e^{-j(\omega-\omega_0)N}}{1-e^{-j(\omega-\omega_0)}} + \frac{1-e^{-j(\omega+\omega_0)N}}{1-e^{-j(\omega+\omega_0)}}\right]$$

$$= \frac{A}{2}\left[\left(\frac{e^{-j\frac{\omega-\omega_0}{2}N}}{e^{-j\frac{\omega-\omega_0}{2}}}\right)\left(\frac{e^{j\frac{\omega-\omega_0}{2}N} - e^{-j\frac{\omega-\omega_0}{2}N}}{e^{j\frac{\omega-\omega_0}{2}} - e^{-j\frac{\omega-\omega_0}{2}}}\right) + \left(\frac{e^{-j\frac{\omega+\omega_0}{2}N}}{e^{-j\frac{\omega+\omega_0}{2}}}\right)\left(\frac{e^{j\frac{\omega+\omega_0}{2}N} - e^{-j\frac{\omega+\omega_0}{2}N}}{e^{j\frac{\omega+\omega_0}{2}} - e^{-j\frac{\omega+\omega_0}{2}}}\right)\right]$$

$$X_d(\omega) = \frac{A}{2}\left[e^{-j\frac{\omega-\omega_0}{2}(N-1)}\left(\frac{\sin\left(\frac{N}{2}(\omega-\omega_0)\right)}{\sin\left(\frac{1}{2}(\omega-\omega_0)\right)}\right) + e^{-j\frac{\omega+\omega_0}{2}(N-1)}\left(\frac{\sin\left(\frac{N}{2}(\omega+\omega_0)\right)}{\sin\left(\frac{1}{2}(\omega+\omega_0)\right)}\right)\right]$$

$C(\omega-\omega_0)$ $C(\omega+\omega_0)$

DTFT of Finite-length Sinusoid

Let $x[n] = A\cos(\omega_0 n),\ 0 \le n \le N-1.$

Then,

$$X_d(\omega) = \frac{A}{2}C(\omega-\omega_0) + \frac{A}{2}C(\omega+\omega_0),$$

where

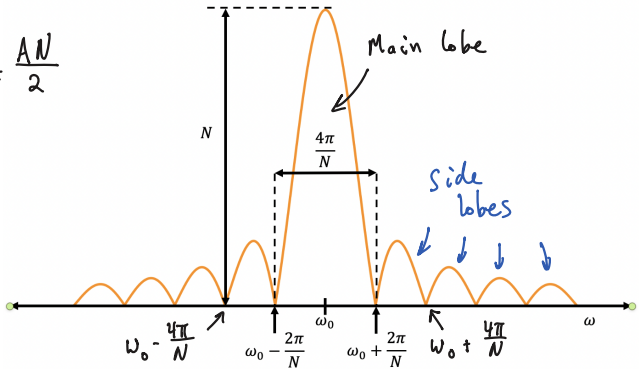
$$C(\omega) = e^{-j\frac{\omega}{2}(N-1)}\left(\frac{\sin\left(\frac{N}{2}\omega\right)}{\sin\left(\frac{1}{2}\omega\right)}\right)$$

→ Periodic or digital sinc

→ Dirichlet function

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Overall height of main lobe = $\frac{AN}{2}$



Magnitude spectrum of $C(\omega-\omega_0)$.

Example 1: DTFT of Finite-length Sinusoidal Signals

Let $x[n] = A\cos(\omega_0 n),\ 0 \le n \le N-1$. Part of the magnitude spectrum of $x[n]$, $|X_d(\omega)|$ is shown below. Determine the frequency ω_0, length N, and amplitude A of $x[n]$.

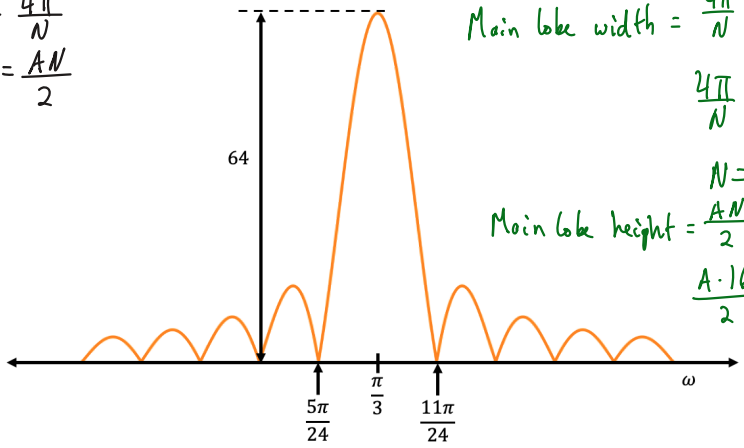
Main lobe width $= \frac{4\pi}{N}$

Main lobe height $= \frac{AN}{2}$

$\omega_0 = \frac{\pi}{3}$

$N = 16$

$A = 8$



Main lobe width $= \frac{4\pi}{N} = \frac{11\pi}{24} - \frac{5\pi}{24}$

$\frac{4\pi}{N} = \frac{6\pi}{24} = \frac{\pi}{4}$

$N = 16$ samples

Main lobe height $= \frac{AN}{2} = 64$

$\frac{A \cdot 16}{2} = 64$

DFT of Finite-length Sinusoid

The DFT of a finite-length sinusoidal signal may be quickly obtained via

$$X[k] = X_d\left(\frac{2\pi k}{N}\right),\ 0 \le k \le N-1.$$

Thus, for $x[n] = A\cos(\omega_0 n),\ 0 \le n \le N-1$,

$$X_d(\omega) = \frac{A}{2}C(\omega - \omega_0) + \frac{A}{2}C(\omega + \omega_0)$$

$$X[k] = X_d(\omega) = \frac{A}{2}C\left(\frac{2\pi k}{N} - \omega_0\right) + \frac{A}{2}C\left(\frac{2\pi k}{N} + \omega_0\right),\ 0 \le k \le N-1.$$

However, the length of $x[n]$ plays a critical role in how $X[k]$ looks.



DFT of Finite-length Sinusoid

Consider two signals:

$$x_1[n] = \cos\left(\frac{\pi}{3}n\right), 0 \le n \le 23$$

$$x_2[n] = \cos\left(\frac{\pi}{3}n\right), 0 \le n \le 21.$$

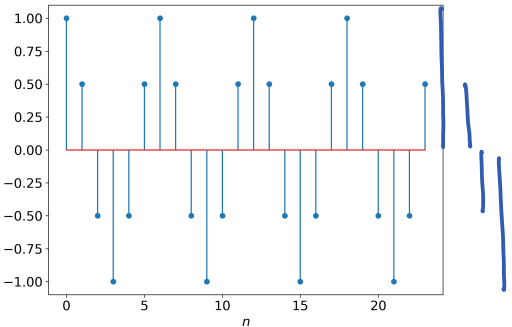
4 cycles of $\cos(\frac{\pi}{3}n)$ → $N_1 = 24$

3.66 cycles of cos → $N_2 = 22$

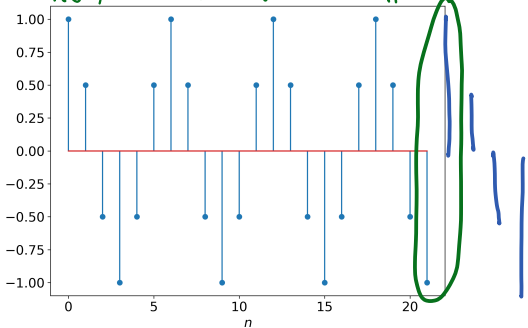
Referred to as "spectral leakage"

For what values of k will $X_1[k]$ and $X_2[k]$ be maximized?

This discontinuity due to periodic extension causes new, erroneous frequencies to appear in $X[k]$



(a) $x_1[n]$



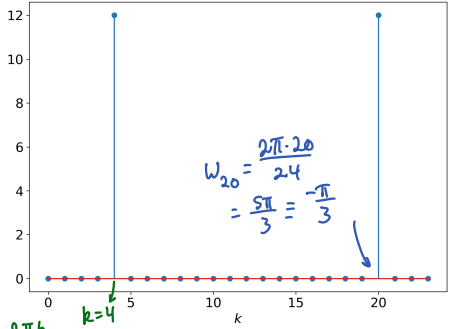
(b) $x_2[n]$

DFT of Finite-length Sinusoid

Consider two signals:

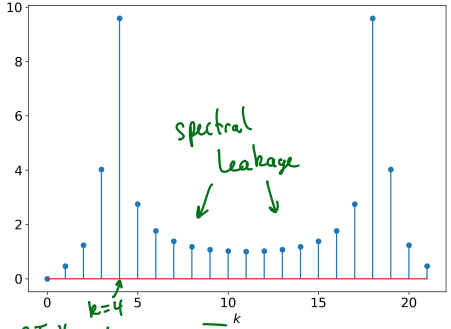
$$x_1[n] = \cos\left(\frac{\pi}{3}n\right), 0 \le n \le 23$$

$$x_2[n] = \cos\left(\frac{\pi}{3}n\right), 0 \le n \le 21.$$



$W_4 = \frac{2\pi k}{N} = \frac{2\pi \cdot 4}{24} = \frac{\pi}{3}$

(a) $X_1[k]$



$W_4 = \frac{2\pi \cdot 4}{22} = \frac{4\pi}{11} \approx 0.36\overline{36}\,\pi$

(b) $X_2[k]$

Example 2: DFT of Finite-length Sinusoidal Signals

For each of the following signals, determine if their DFT will have exactly two or four non-zero values or more than two/four non-zero values.

(a) $x_1[n] = \sin\left(\frac{\pi}{11}n\right), \ 0 \leq n \leq 32.$

(b) $x_2[n] = \sin\left(\frac{3\pi}{5}n\right), \ 0 \leq n \leq 19.$

(c) $x_3[n] = \cos\left(\frac{\pi}{4}n\right) + \cos\left(\frac{\pi}{3}n\right), \ 0 \leq n \leq 47.$



Key Spectral Analysis Challenges

There are multiple possible challenges with performing spectral analysis. Fortunately, we can develop the same useful tools via the simpler case of $M = 2$ components.

$$s[n] = A_1 \cos(\omega_1 n) + A_2 \cos(\omega_2 n), \ 0 \leq n \leq N - 1.$$

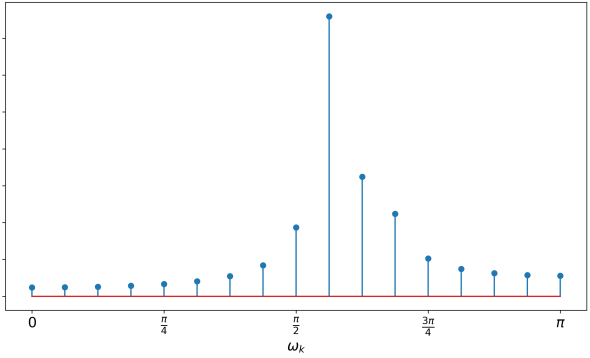


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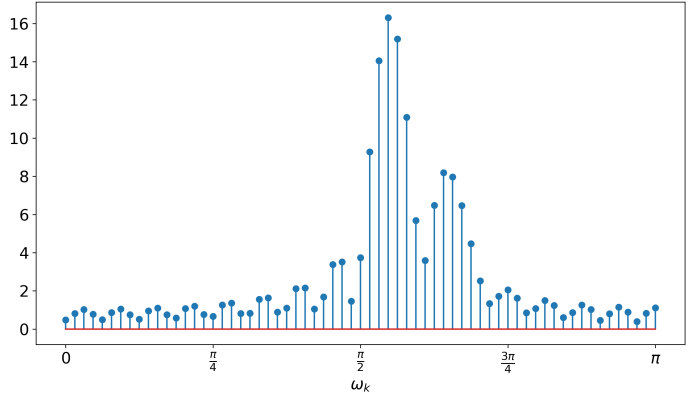
Consider the case where $A_1 = 1$, $A_2 = \frac{1}{2}$, $\omega_1 = 0.55\pi$, $\omega_2 = 0.65\pi$, $N = 32$.



Magnitude spectrum of $S[k]$ for $0 \leq k \leq 15$.

Key Spectral Analysis Challenges

In this case, zero-padding $s[n]$ from length–32 to length–128 is sufficient to get a good picture of each spectral component. Let $\tilde{s}[n]$ denote the zero-padded version of the signal from the previous slide.



Magnitude spectrum of $\tilde{S}[k]$ for $0 \leq k \leq 63$.

Key Spectral Analysis Challenges

There are two key challenges we will examine in performing spectral analysis. Once again, let $s[n] = A_1 \cos(\omega_1 n) + A_2 \cos(\omega_2 n)$.

1. **The frequencies of components may be too close to separate.**

   If $|\omega_2 - \omega_1|$ is too small, then the main lobes of each component overlap.

2. **The amplitudes of components may vary greatly.**

   If $A_1 \gg A_2$, then the side lobes of component 1 will be taller than the main lobe of component 2!

   ($\gg$: "much greater")

   E.g. $A_1 = 1$, $A_2 = \frac{1}{100}$